

Stress Testing Prospective Multimodal Dataset Utility Prediction

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ABSTRACT

Before choosing how to train a multimodal model, a practitioner often faces a more basic expense: deciding which corpus deserves a training run. We study how much of that decision can be made from frozen representations before any candidate is added to the target model. Our sealed benchmark hides transfer outcomes and target validation labels until every score is fixed. It covers three sentiment and emotion corpora, five seeds, three repartition families, and four downstream learners. For logistic regression, centroid distance routes to the oracle source in 44 of 45 decisions. The signal is genuinely coarse: on a held out family, the gap between sources is 22.3 times the typical variation among buckets from one source, while equal size and class balanced controls leave within source rank correlation near zero. Learner choice changes the target of selection. GaussianNB gains from text candidates but loses from every audio candidate, and matched fusion ablations show route accuracy falling from 11 of 15 with fixed weights to 8 of 15 with a sample dependent gate. Canonical LEEP and LogME do not directly score additive data buckets, so we evaluate candidate trained adaptations. Neither is reliable as a flat selector. A practical hybrid works better: distance first chooses a source, adapted LEEP shortlists two of sixteen buckets, and exact pilots among those two yield mean utility 0.00971 while avoiding 87.5% of full candidate runs. The evidence supports prospective screening as a staged budget allocation procedure, not as a universal dataset quality score.

1 INTRODUCTION

Suppose an audio and text project can access many corpora but afford only a few full transfer runs. Before comparing architectures, the team must decide which data deserve those runs. A candidate source may improve the target model, leave it unchanged, or make it worse by shifting label and feature distributions. Training on every candidate would answer the question only after spending the budget that selection was meant to save.

This decision is not captured by data quality alone. Clean labels and aligned modalities remain desirable, yet neither establishes that a source will help a particular target and learner. We therefore define utility as $\mathcal{U}(\mathcal{S}; \mathcal{T}, M, R)$: the source $\mathcal{S}$, target $\mathcal{T}$, model M , and training recipe R jointly determine the quantity to be predicted. The same corpus can consequently be useful in one downstream setting and harmful in another.

Two distinctions determine whether a prospective score is useful. First, separating distant corpora is easier than ordering similar buckets drawn from one corpus; success at the former need not imply success at the latter. Second, a ranking learned for one downstream learner need not survive a change in model. We test both distinctions under a sealed protocol: scores are evaluated on held out targets or repartitions, and candidate utilities are revealed only after routes, bucket rankings, and abstention decisions have been frozen. Abstention is reported with coverage, since rejecting every candidate is safe but useless.

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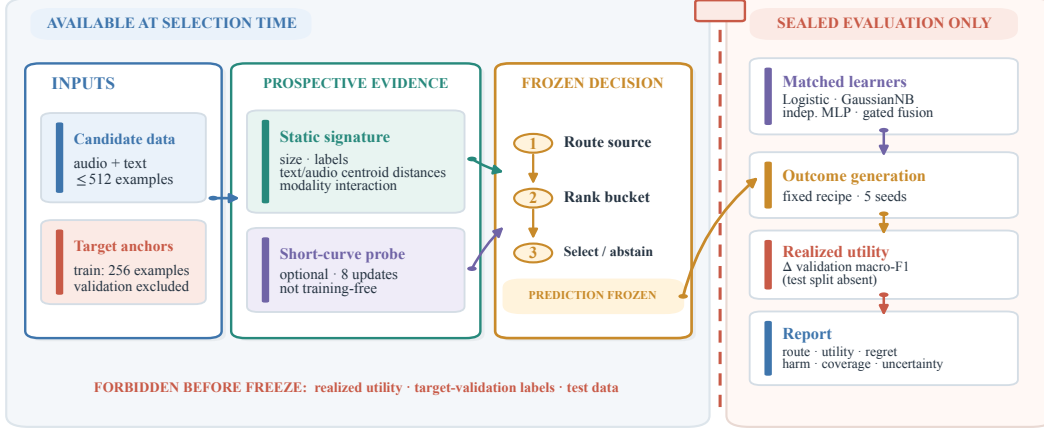


Figure 1: Information boundary and evaluation flow. Candidate data and target training anchors produce static signatures and, when enabled, an eight update probe. The probe is not training free. Routes, bucket rankings, and abstention decisions are frozen before candidate utilities or target validation labels are revealed. Full downstream training produces only the sealed evaluation outcomes.

The paper makes four contributions:

- We define multimodal dataset utility with a strict information boundary and provide a deterministic five seed protocol built on common text and audio representations.
- We isolate and explain a granularity gap. For logistic regression, source routing is stable across repeated partitions (44/45 decisions over three independent target corpora), while bucket ranking and calibrated abstention are not. Split half, equal size, and class balanced controls show that the routing signal reflects a large source scale offset.
- We measure target level uncertainty and repeat the same candidate comparisons with GaussianNB, neural MLP, and learned gated fusion. The matched panels reveal learner dependence that candidate level averages would otherwise conceal. Matched gate and GaussianNB ablations locate two concrete causes: adaptive fusion changes the route, and audio moment shifts erase text gains under GaussianNB.
- We compare candidate trained adaptations of LEEP and LogME under the same seal and turn the negative ranking result into a cost aware workflow: route a source statically, shortlist buckets with a cheap probe, then spend full training only on the shortlist and retain the option to add no source.

2 RELATED WORK

Transferability estimation. LEEP, LogME, and TransRate estimate the transferability of a pretrained model or representation without exhaustive fine tuning (Nguyen et al., 2020; You et al., 2021; Huang et al., 2022). These methods rank source models or feature spaces. We instead hold common representations fixed and rank candidate training data by the change it induces in a specified downstream learner. Canonical LEEP requires the predictions of a source trained classifier on labeled target examples. LogME evaluates the evidence of target labels under a supplied representation. Neither produces a different score for candidate data buckets when the target representation is fixed. We therefore report them both in their original scope and as explicitly labeled candidate trained probes.

Data valuation and task aware selection. Data Shapley assigns value through marginal downstream utility, whereas datamodels learn a mapping from training subsets to model predictions (Ghorbani & Zou, 2019; Ilyas et al., 2022). DsDm uses this model aware view for language model dataset selection, and LESS uses optimizer aware gradient similarity for targeted instruction tuning (Engstrom et al., 2024; Xia et al., 2024). Together, these approaches show that usefulness can depend on the target and learner, but their scores require subset training supervision, warmup training, or gradients. Our static signatures obey a stricter boundary before candidate training; short probes are reported separately rather than described as training free.

Multimodal curation. DataComp standardizes image and text filtering under fixed compute, while JEST selects jointly learnable multimodal batches using learner and reference losses (Gadre et al., 2023; Evans et al., 2024). Recent preprints further combine intrinsic quality, relevance, diversity, validation feedback, capability specific raters, or one step proxy updates (Cheng et al., 2026; Sahi et al., 2026; Jing et al., 2026). They primarily optimize large scale vision and language pretraining or multimodal instruction tuning. Our setting instead asks whether whole audio and text sources and their internal buckets can be screened before running a fixed target augmentation recipe.

Training free supervision data generalization. DataProphet is the closest precedent. It predicts influence rankings for vision and language supervision datasets from perplexity, similarity, and diversity, without candidate specific training (Qi et al., 2026). We do not claim the first training free multimodal selector. Instead, we test how far this premise extends when selection moves from corpora to buckets and when the downstream learner changes. The evaluation adds sealed natural repartitions, target cluster uncertainty, and independently calibrated abstention.

3 PROBLEM FORMULATION

Let $D_{\mathcal{T}}^{\text{tr}}$ and $D_{\mathcal{T}}^{\text{val}}$ denote target training and validation data, and let $\mathcal{S}_i$ be a candidate source block. For a fixed model and recipe (M, R) , we define realized utility under seed s as

$$u_{i,s} = q(M_R(D_{\mathcal{T}}^{\text{tr}} \cup \mathcal{S}_i; s), D_{\mathcal{T}}^{\text{val}}) - q(M_R(D_{\mathcal{T}}^{\text{tr}}; s), D_{\mathcal{T}}^{\text{val}}), \quad (1)$$

where q is macro F1. Ground truth utility is $\mathcal{U}_i = |\mathcal{K}|^{-1} \sum_{s \in \mathcal{K}} u_{i,s}$ over five locked seeds. Positive utility means that adding the candidate improves validation performance on average.

The selector may use candidate data, target training anchors, model metadata, and features derived from them. It may not use candidate specific $u_{i,s}$, target validation labels, or target test data. Given scores $\hat{u}_i$, forced selection chooses $\arg \max_i \hat{u}_i$. We report selected utility and top one regret

$$\text{Regret} = \max_i \mathcal{U}_i - \mathcal{U}_i. \quad (2)$$

A selective rule may abstain, in which case its realized utility is zero. We report its coverage and accepted harm, defined by $\mathcal{U}_i < 0$.

4 EVALUATION FRAMEWORK

4.1 DATA AND COMMON REPRESENTATIONS

The multimodal panel comprises CMU-MOSI (Zadeh et al., 2016), MELD (Poria et al., 2019), and CH-SIMS (Yu et al., 2020). Only audited training and validation splits enter the study. We map sentiment labels to negative, neutral, and positive classes. Text is represented by the frozen 384 dimensional paraphrase-multilingual-MiniLM-L12-v2 sentence transformer (Reimers & Gurevych, 2019; Wang et al., 2020); audio is encoded by the frozen 384 dimensional whisper-tiny encoder (Radford et al., 2022). Model revisions are fixed in the released manifests, and candidate and target IDs are aligned before transfer experiments. Target anchors contain 256 examples and candidate blocks contain at most 512 examples.

The development partition hashes video or dialogue identifiers and therefore keeps groups intact. The three confirmation families instead hash salted sample identifiers. They are deterministic, but they do not preserve groups. Source examples can move between salts and reappear in another family, so salts and families are repeated measurements rather than independent datasets.

For each target, 256 training anchors define the baseline. Candidate data are appended with total source weight capped at the anchor weight. Separate text and audio heads produce three class probabilities, followed by equal weight fusion. Utility is the change in validation macro F1 from the anchor only baseline, averaged over five fixed seeds.

4.2 PROSPECTIVE SIGNATURES AND HIERARCHICAL DECISIONS

The static signature records candidate size, label statistics, target to source centroid distances for text and audio, and cross modal interaction descriptors. An eight update probe adds learning curve features from support and query splits of the target training data. The probe never accesses target validation data or a candidate’s realized utility, but it is not training free.

Selection proceeds in two stages. Source routing chooses the corpus with the smallest mean text/audio centroid distance across its eight buckets. Within that source, Curve 19 ranks buckets by Ridge regression on the 19 dimensional probe signature. A balanced logistic classifier provides a second score by predicting $\mathcal{U}_i > 0$. Both selectors are fit in leave one target out folds. Standardization within a target and source stratum uses candidate features but no utilities.

4.3 CANDIDATE TRAINED TRANSFERABILITY PROBES

We place LEEP and LogME behind the same seal, with one necessary change of scope. For each candidate bucket, class balanced logistic heads are fitted to its frozen text and audio representations. Their aligned probabilities on the labeled target anchor are averaged for adapted LEEP. Source head LogME uses the concatenated text and audio log probabilities as a six dimensional target representation. Scores are averaged across five anchor seeds. These methods fit a small model to every candidate and are therefore probe baselines, not static selectors. We report flat ranking and ranking after the frozen distance route.

4.4 SEALED CONFIRMATION AND UNCERTAINTY

Each of the three separately salted families contains 240 candidates: three targets, two eligible sources, five salts, and eight buckets. The families are separated by protocol but not disjoint by sample. Prediction artifacts are written before utility generation. Rejection thresholds are chosen on a dedicated calibration family and evaluated once on the final family.

Partition decisions share corpora and are not independent replicates. For primary uncertainty estimates, effects are first averaged within targets and then entered into a hierarchical bootstrap with 10,000 draws over targets, batches, and salts. We also enumerate all 2^3 target level sign flips. Three target clusters cannot support exact significance at 0.05, and salt level tests would overstate the effective sample size.

4.5 DOWNSTREAM LEARNERS

The primary recipe uses class balanced logistic regression heads for text and audio. Matched panels replace both heads with GaussianNB or independent two layer MLPs while holding candidates, weights, representations, seeds, and the validation metric fixed. Each MLP has a 64 unit ReLU hidden layer and receives 80 deterministic, full batch AdamW updates. A fourth panel learns fusion jointly: modality specific 384 to 64 ReLU branches feed a two way softmax gate and a shared three class head, again trained for 80 fixed updates. To isolate the effect of gating, matched variants use fixed equal weights, one learned global weight, or the original sample dependent gate. All other parameters and updates are held constant.

Table 1: Audited computation. Counts are exact; wall times are local CPU measurements on the same panel of 240 candidates and are not estimates of neural model cost. Each fused evaluation fits separate text and audio heads.

Stage	Blocks	Primitive computation	Wall time (s)
Curve 19 screening	240	9,600 fused updates	14.46
One logistic utility panel	240	1,200 seeded evaluations	17.88
Three logistic panels	720	7,290 head fits	N/A
Matched GaussianNB panel	240	2,430 head fits	N/A
Matched neural MLP panel	240	2,430 head fits	83.55
Complete reported audit	1,200	12,150 head fits	N/A

GaussianNB, independent MLP, and gated fusion anchor macro F1 ranges are 0.442 to 0.532, 0.437 to 0.540, and 0.423 to 0.543, respectively, ruling out completely degenerate comparisons.

4.6 COMPUTE ACCOUNTING

Table 1 separates exact training counts from machine dependent timings. The three sealed logistic families and the matched GaussianNB and MLP panels contain 6,000 candidate and seed evaluations, requiring 12,150 fitted modality heads after shared anchor baselines are included. The gated fusion panel adds 1,200 candidate and seed evaluations and 1,215 jointly trained models, or 97,200 full batch updates. In a deterministic rerun of one panel with 240 candidates, Curve 19 screening takes 14.46 seconds, compared with 17.88 seconds for full logistic utility generation. The 19.1% wall time reduction is modest because both methods are inexpensive in this classical setting. All 1,200 stored utility rows are reproduced exactly.

For the motivating choice of 10 among 100 candidates, exhaustive five seed evaluation with two heads requires 1,000 candidate head fits. Evaluating ten selected candidates requires 100, avoiding 90% of candidate fits, or 89.1% after a shared ten fit baseline is included. This calculation is conditional; it is neither an observed speedup nor evidence that the selected ten are optimal. End to end neural savings remain unmeasured.

5 RESULTS

5.1 SOURCE ROUTING IS EASIER THAN BUCKET SELECTION

The hierarchy localizes the main error. Routing followed by the original bucket score lowers regret from 0.04278 to 0.02730, yet mean utility remains negative (Table 2). An oracle restricted to the routed source attains positive utility on all three targets with regret 0.00060. Under the logistic recipe, useful buckets are present in the routed source; the selector fails to identify them reliably.

The failure is not explained by bucket size or class proportion. On the sealed safety family, the median distance gap between two sources is 22.33 times the standard deviation among buckets from one source. Independent halves choose the same source in all 15 decisions. Inside a source, split half distance reliability is $\tau = -0.152$. Its utility correlation is $\tau = 0.005$, and falls to -0.010 after equal size subsampling and -0.038 after class balanced centroids. Distance therefore measures a reproducible domain offset. The smaller differences among nearby buckets are neither reliable nor aligned with utility.

5.2 APPARENT GAINS REVERSE ACROSS REPARTITIONS

On the first sealed family, the positive utility classifier lowers regret from 0.01780 to 0.01211 relative to Curve 19 and selects a beneficial bucket in 10 of 15 decisions. The next two families reverse the gain, with utility differences of -0.00153 and -0.00102 (Figure 2).

Table 2: Hierarchical error decomposition on the initial three target panel. Oracle diagnostics are not deployable. The small routed source oracle regret isolates bucket discrimination within a source as the logistic model bottleneck.

Decision rule	Mean utility $\uparrow$	Regret $\downarrow$	Positive targets
Flat original 12	-0.02800	0.04278	0/3
Hierarchical original 12	-0.01252	0.02730	1/3
Routed source oracle bucket	+0.01417	0.00060	3/3

Table 3: Mechanism controls and the staged selection workflow on the sealed safety family. Utility is the change in macro F1. Route accuracy is measured over 15 target and partition decisions.

Analysis	Result	Control or cost
Distance scale	source gap / bucket spread 22.33	split half route 15/15
Bucket rank	raw / equal size / balanced τ : .005 / -.010 / -.038	split half $\tau = -.152$
GaussianNB	positive text / audio / fusion: 54.2 / 0.0 / 3.75%	smoothing 3.3 to 5.0%
Gate route	fixed / global / sample: 11 / 10 / 8 of 15	matched network and updates
Adapted LEEP	flat / routed utility: -.01540 / .00497	harm 66.7 / 46.7%
Source head LogME	flat / routed utility: -.00350 / -.00092	harm 53.3 / 53.3%
LEEP shortlist	exact pilots $k = 2$ / $k = 4$: .00971 / .01435	runs saved 87.5 / 75.0%

Across all 45 decisions, the mean difference is 0.001047; its 95% hierarchical interval is $[-0.003804, 0.005616]$, and the exact target level test gives $p = 0.50$.

A later baseline audit reaches the same cautionary conclusion. Uniform bucket selection has expected utility 0.00077 and regret 0.01810, compared with 0.00178/0.01727 for the classifier and 0.00073/0.01832 for Curve 19 (Table 4). Curve 19 does not outperform random selection on these aggregate metrics. Because the random baseline was added after the primary outcomes were inspected, the classifier’s margin over it remains descriptive.

5.3 CLASSIFIER CONFIDENCE DOES NOT SUPPORT SAFE ABSTENTION

The calibration family yields no eligible probability threshold. Between 0.50 and 0.99, accepted harm ranges from 50% to 75%, and realized utility is negative at every grid point (Figure 3). The locked fallback threshold of 1.0 rejects every final decision. It prevents accepted harm only by reducing both coverage and realized utility to zero.

5.4 UTILITY RANKINGS DEPEND ON THE DOWNSTREAM LEARNER

Matched logistic regression and GaussianNB utilities agree moderately in rank, but not in sign (Figure 4). Beneficial candidates account for 28.75% of the logistic panel and only 3.75% of the GaussianNB panel. On the same family, route accuracy falls from 14/15 to 10/15. Both frozen bucket selectors choose harmful GaussianNB candidates in all 15 decisions (Table 4).

The GaussianNB collapse comes from the audio branch rather than numerical smoothing. Across variance smoothing values from 10^{-11} to 10^{-3} , only 3.3% to 5.0% of fused candidates are beneficial. At the locked value, 54.2% of text candidates help, but no audio candidate does. Mean audio utility is -0.0884 , -0.0806 , and -0.0792 on the three targets. Fusion inherits that loss even when text improves. Class conditional audio variance shift is negatively associated with audio utility ($\rho = -0.448$), while candidate size is almost unrelated to audio utility ($\rho = 0.016$). These controls do not prove that Gaussian independence assumptions are the only cause, but they rule out a single unlucky smoothing value and locate the loss in cross corpus audio moment updates.

The MLP panel preserves some coarse structure but little within source agreement. The distance route matches the oracle source in 12/15 decisions. Within a routed source, logistic to MLP Kendall agreement is only $\tau = 0.1619$, compared with candidate level $\tau = 0.3183$ and 67.92% sign agreement. The frozen positive utility classifier attains mean utility

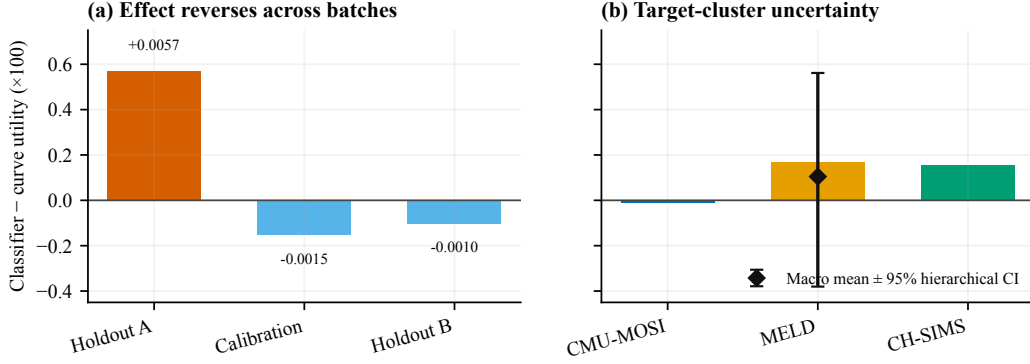


Figure 2: Selector gains depend on the repartition family. (a) The positive utility classifier exceeds Curve 19 on the first sealed family, then reverses on the calibration and final families. (b) Target level effects and the hierarchical bootstrap interval, which includes zero.

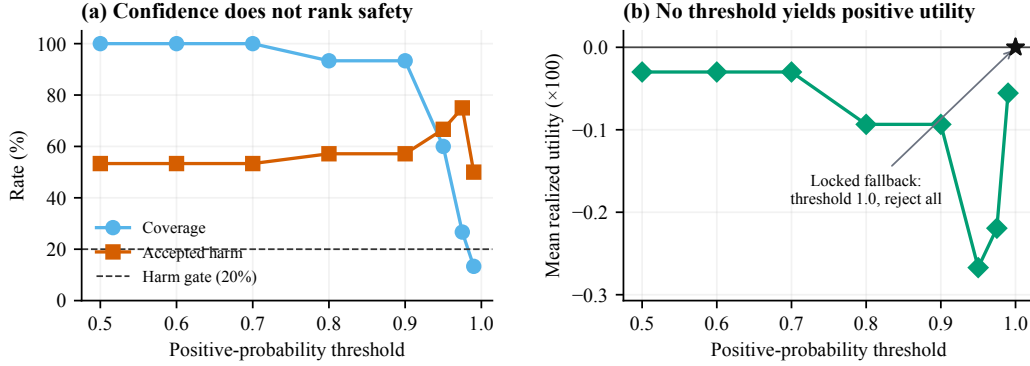


Figure 3: Classifier probability does not provide a useful safety threshold. Raising the positive class threshold reduces coverage without controlling accepted harm or producing positive utility. The locked fallback rejects every candidate.

+0.00755 with 26.7% harm, but the corresponding target means are +0.02754, -0.00356, and -0.00132. Its aggregate gain is therefore driven by one target. Two CPU runs reproduce all 1,200 utilities exactly in 83.55 and 82.56 seconds.

The matched gate ablation explains why learned fusion disrupts routing. With fixed equal weights, the distance route matches the oracle source in 11/15 decisions. A learned global gate lowers this to 10/15, and a sample dependent gate lowers it again to 8/15. For the frozen positive utility selector, mean selected utility moves from +0.00511 to +0.00182 and then -0.00027; harm rises from 26.7% to 53.3% at the final step. Mean absolute gate shift from the anchor value grows from zero to 0.0074 and 0.0901. Because the branches, shared head, optimizer, and update count are identical, architecture size and training budget cannot explain this sequence. The evidence instead points to objective drift: an adaptive gate changes how each source perturbs the trained decision rule, while the frozen route still measures geometry before that adaptation. Both encoders remain frozen, so the result does not cover end to end tuning.

5.5 TRANSFERABILITY BASELINES AND A PRACTICAL COMPROMISE

Canonical LEEP and LogME cannot distinguish our buckets without a candidate specific source model. Their candidate trained adaptations remain weak when used flat. Adapted LEEP has source stratified $\tau = 0.098$, mean selected utility -0.01540, and 66.7% harm.

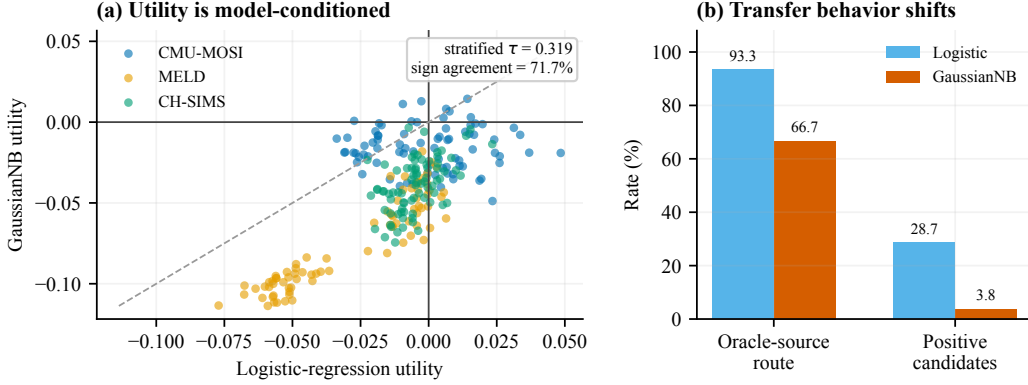


Figure 4: The downstream learner changes utility and routing. (a) Candidate utilities retain moderate rank agreement but shift downward under GaussianNB. (b) Oracle source routing and the proportion of beneficial candidates both decline on the matched panel.

Table 4: Frozen selector decisions across downstream learners. Logistic results pool three sealed partition families (45 decisions); GaussianNB, the independent head neural MLP, and learned gated fusion use the matched final family (15 decisions each). Utility is validation macro F1 change. Route is oracle source accuracy and is shared by methods within a model. Expected random and size only results ($\dagger$) average ties analytically and are descriptive baselines added after the primary analysis.

Model	Bucket rule	Utility $\uparrow$	Positive % $\uparrow$	Harm % $\downarrow$	Regret $\downarrow$	Route % $\uparrow$
Logistic	Positive utility classifier	+0.00178	51.1	48.9	0.01727	97.8
Logistic	Curve 19	+0.00073	40.0	60.0	0.01832	97.8
Logistic	Uniform random $\dagger$	+0.00077	43.1	56.9	0.01810	97.8
Logistic	Size only $\dagger$	-0.00050	40.8	59.2	0.01937	97.8
GaussianNB	Positive utility classifier	-0.03734	0.0	100.0	0.02773	66.7
GaussianNB	Curve 19	-0.03744	0.0	100.0	0.02783	66.7
Neural MLP	Positive utility classifier	+0.00755	73.3	26.7	0.01405	80.0
Neural MLP	Curve 19	+0.00486	66.7	33.3	0.01674	80.0
Gated fusion	Positive utility classifier	-0.00027	46.7	53.3	0.02075	53.3
Gated fusion	Curve 19	-0.00413	26.7	73.3	0.02460	53.3

Source head LogME obtains $\tau = 0.119$, utility -0.00350 , and 53.3% harm. Thus our static bucket scores do not establish a general advantage over these baselines. After distance routing, adapted LEEP improves to $+0.00497$ mean utility, with 46.7% harm, whereas LogME remains negative. The route and the probe solve different parts of the decision.

This observation yields a budgeted procedure. First use distance only to choose a source. Next use adapted LEEP to rank its eight buckets. Finally run the exact downstream recipe on a small shortlist and keep the anchor model when every pilot is harmful. With $k = 2$, full training is required for 2 of 16 candidates. The best pilot has mean utility $+0.00971$ with a decision bootstrap interval of $[0.00177, 0.01827]$, while 87.5% of full candidate runs are avoided. With $k = 4$, mean utility rises to $+0.01435$ and 75% of runs are avoided. The last stage is deliberately not training free. It converts an unreliable one shot ranking problem into a smaller empirical choice and makes the remaining cost and risk explicit.

5.6 A FOURTH CORPUS DOES NOT RESCUE TEXT ONLY BUCKET RANKING

An auxiliary text only experiment adds CMU-MOSEI (Zadeh et al., 2018) as a fourth independent target and source. This panel separates the small number of targets from the availability of aligned audio and contains 96 target and candidate pairs, or 480 utility evaluations across five seeds. The full text signature reaches target level Kendall $\tau = 0.1486$, with a 95% target cluster interval of $[-0.1884, 0.4855]$ (Table 6). Text centroid distance has

Table 5: Matched utility diagnostics across models over 240 candidates, using logistic regression as reference. Candidate level agreement overstates the stability within a stratum that matters for bucket selection.

Diagnostic	GaussianNB	Neural MLP	Gated fusion
Candidate Kendall τ	0.4222	0.3183	0.3503
Source stratified Kendall τ	0.3190	0.1619	0.1452
Utility sign agreement	71.67%	67.92%	63.75%
Positive candidates	3.75%	51.67%	45.00%
Positive candidates, logistic		28.75%	

Table 6: Auxiliary four corpus text only panel with one target held out, adding CMU-MOSEI as a fourth independent target. Brackets are exploratory 95% target cluster bootstrap intervals over only four targets. Lower regret is better.

Rule	Target Kendall τ	Top one regret	Within source τ
Size only	-0.3233 [-0.5695, -0.0764]	0.03573 [0.02466, 0.04681]	-0.0952
Length only	+0.1449 [-0.0797, +0.3043]	0.02255 [0.01098, 0.03766]	+0.0536
Text centroid only	+0.2572 [+0.0362, +0.4783]	0.02183 [0.00801, 0.03564]	-0.1369
Full text signature	+0.1486 [-0.1884, +0.4855]	0.00583 [0.00180, 0.01022]	-0.1369

a higher point estimate, $\tau = 0.2572$, whereas the full signature is negatively associated with utility within source, $\tau = -0.1369$. This experiment is not multimodal and cannot strengthen the main scope, but adding one target does not resolve the ranking problem within a source.

6 DISCUSSION AND LIMITATIONS

The experiments support a narrower use of prospective scores than a single leaderboard rank suggests. Domain proximity can eliminate an implausible source, but should not choose among nearby buckets. Classifier confidence should not be used as a safety certificate because it reduces coverage without controlling harm. A practitioner should name the downstream learner first, use static geometry only for coarse routing, and reserve a small validation budget for candidate trained probes or exact pilots. Keeping the anchor only model as an explicit option prevents a forced harmful addition after those pilots. The choice of shortlist size then exposes a concrete tradeoff: two pilots avoid 87.5% of candidate runs, while four pilots avoid 75% and recover more utility.

Several limitations bound these findings. The multimodal study has three independent target corpora, all drawn from sentiment or emotion tasks. Every panel uses frozen representations. The MLP broadens the architecture check, and gated fusion learns modality weights, but neither substitutes for end to end encoder fine tuning. Utility is validation macro F1 under one augmentation recipe; changing the source weight, metric, fusion rule, or training budget defines a new quantity. Finally, the natural repartitions reuse source examples across salts. They diversify decisions but do not create additional independent corpora.

The present evidence also does not justify a claim about images or geospatial time series. A common frozen image encoder could be added to these corpora with sparse frame sampling, but that would test a third modality within the same affective domain. Geospatial data introduce different prediction units, labels, and temporal dependence, so they require a new benchmark rather than another feature column. Both extensions should preserve the same seal and repeat the fixed, global, and sample dependent fusion controls before any general claim is made.

7 CONCLUSION

Prospective multimodal data selection must specify the downstream learner and the granularity of the decision. Corpus routing is reliable for the logistic recipe, but it does not become a reliable bucket score. GaussianNB and matched gate ablations explain two ways in which the target changes across learners. The resulting recommendation is concrete: route statically, probe within the routed source, train only a short list, and retain the option to add nothing. General claims will require more independent targets, new domains, and end to end models; current evidence supports a transparent cost and risk tradeoff instead.

AI USE STATEMENT

Generative AI assisted with hypothesis and methodology refinement, literature discovery, translation, code and figure drafting, data artifact formatting, result interpretation, and manuscript editing. It was not used to create training or evaluation examples, ground truth labels, mathematical claims, or proofs. The authors checked citations against primary records, executed and tested generated code locally, and verified reported numbers against versioned machine readable artifacts. The authors reviewed all assisted work and take responsibility for the final text, claims, code, and artifacts.

ETHICS STATEMENT

This work conducts secondary analysis of existing multimodal sentiment and emotion corpora and recruits no new participants. Raw speech, video, and transcripts remain subject to the access conditions and licenses of the original corpora; supplementary materials will distribute only code and derived artifacts permitted by those terms. Affective labels and representations can encode demographic, linguistic, cultural, and annotator biases. Dataset utility rankings should therefore not be interpreted as judgments about the quality or worth of speakers, nor used to justify high stakes affect inference. Our conclusions are limited to the audited benchmark tasks, representations, models, metrics, and training recipes.

REPRODUCIBILITY STATEMENT

Section 3 specifies the problem and information boundary. Section 4 records the datasets, representations, signatures, partitions, seeds, downstream recipes, decision rules, uncertainty analysis, and compute accounting. Figures and tables are generated from versioned machine readable artifacts. Subject to corpus licenses, the anonymized supplement will include code for data audits, utility panels, selectors, statistical analyses, tables, and figures, together with artifact hashes.

REFERENCES

- Guanjie Cheng, Boyi Li, Lingyu Sun, Mengying Zhu, Yangyang Wu, Xinkui Zhao, and Shuiguang Deng. TADS: Task-aware data selection for multi-task multimodal pre-training. arXiv preprint arXiv:2602.05251, 2026. URL <https://arxiv.org/abs/2602.05251>. Preprint.
- Logan Engstrom, Axel Feldmann, and Aleksander Madry. DsDm: Model-aware dataset selection with datamodels. In Proceedings of the 41st International Conference on Machine Learning, volume 235 of Proceedings of Machine Learning Research, pp. 12491–12526, 2024.
- Talfan Evans, Nikhil Parthasarathy, Hamza Merzić, and Olivier J. Hénaff. Data curation via joint example selection further accelerates multimodal learning. In Advances in Neural Information Processing Systems, volume 37, 2024. doi: 10.52202/079017-4485.
- Samir Yitzhak Gadre, Gabriel Ilharco, Alex Fang, Jonathan Hayase, Georgios Smyrnis, Thao Nguyen, Ryan Marten, Mitchell Wortsman, Dhruva Ghosh, Jieyu Zhang, et al. DataComp: In search of the next generation of multimodal datasets. In Advances in Neural Information Processing Systems, volume 36, 2023. doi: 10.52202/075280-1179.

- Amirata Ghorbani and James Zou. Data shapley: Equitable valuation of data for machine learning. In Proceedings of the 36th International Conference on Machine Learning, volume 97 of Proceedings of Machine Learning Research, pp. 2242–2251, 2019.
- Long-Kai Huang, Junzhou Huang, Yu Rong, Qiang Yang, and Ying Wei. Frustratingly easy transferability estimation. In Proceedings of the 39th International Conference on Machine Learning, volume 162 of Proceedings of Machine Learning Research, pp. 9201–9225, 2022.
- Andrew Ilyas, Sung Min Park, Logan Engstrom, Guillaume Leclerc, and Aleksander Madry. Datamodels: Predicting predictions from training data. In Proceedings of the 39th International Conference on Machine Learning, volume 162 of Proceedings of Machine Learning Research, pp. 9525–9587, 2022.
- Jinhao Jing, Qiannian Zhao, Chao Huang, and Zhan Su. Efficient data selection for multimodal models via incremental optimization utility. arXiv preprint arXiv:2605.07488, 2026. URL <https://arxiv.org/abs/2605.07488>. Preprint.
- Cuong V. Nguyen, Tal Hassner, Matthias Seeger, and Cédric Archambeau. LEEP: A new measure to evaluate transferability of learned representations. In Proceedings of the 37th International Conference on Machine Learning, volume 119 of Proceedings of Machine Learning Research, pp. 7294–7305, 2020.
- Soujanya Poria, Devamanyu Hazarika, Navonil Majumder, Gautam Naik, Erik Cambria, and Rada Mihalcea. MELD: A multimodal multi-party dataset for emotion recognition in conversations. In Proceedings of the 57th Annual Meeting of the Association for Computational Linguistics, pp. 527–536, 2019. doi: 10.18653/v1/P19-1050.
- Xuan Qi, Luxi He, Dan Roth, and Xingyu Fu. Demystifying supervision data generalization in multimodal LMs. In International Conference on Learning Representations, 2026.
- Alec Radford, Jong Wook Kim, Tao Xu, Greg Brockman, Christine McLeavey, and Ilya Sutskever. Robust speech recognition via large-scale weak supervision. arXiv preprint arXiv:2212.04356, 2022. URL <https://arxiv.org/abs/2212.04356>.
- Nils Reimers and Iryna Gurevych. Sentence-BERT: Sentence embeddings using siamese BERT-networks. In Proceedings of the 2019 Conference on Empirical Methods in Natural Language Processing and the 9th International Joint Conference on Natural Language Processing, pp. 3982–3992, 2019. doi: 10.18653/v1/D19-1410.
- Naveen Sahi, Jeremy Dohmann, Armen Aghajanyan, and Akshat Shrivastava. SkillRater: Untangling capabilities in multimodal data. arXiv preprint arXiv:2602.11615, 2026. URL <https://arxiv.org/abs/2602.11615>. Preprint.
- Wenhui Wang, Furu Wei, Li Dong, Hangbo Bao, Nan Yang, and Ming Zhou. MiniLM: Deep self-attention distillation for task-agnostic compression of pre-trained transformers. In Advances in Neural Information Processing Systems, volume 33, pp. 5776–5788, 2020.
- Mengzhou Xia, Sathika Malladi, Suchin Gururangan, Sanjeev Arora, and Danqi Chen. LESS: Selecting influential data for targeted instruction tuning. In Proceedings of the 41st International Conference on Machine Learning, volume 235 of Proceedings of Machine Learning Research, pp. 54104–54132, 2024.
- Kaichao You, Yong Liu, Jianmin Wang, and Mingsheng Long. LogME: Practical assessment of pre-trained models for transfer learning. In Proceedings of the 38th International Conference on Machine Learning, volume 139 of Proceedings of Machine Learning Research, pp. 12133–12143, 2021.
- Wenmeng Yu, Hua Xu, Ziqi Yuan, and Jiele Wu. CH-SIMS: A chinese multimodal sentiment analysis dataset with fine-grained annotation of modality. In Proceedings of the 58th Annual Meeting of the Association for Computational Linguistics, pp. 3718–3727, 2020. doi: 10.18653/v1/2020.acl-main.343.

Amir Zadeh, Rowan Zellers, Eli Pincus, and Louis-Philippe Morency. Multimodal sentiment intensity analysis in videos: Facial gestures and verbal messages. *IEEE Intelligent Systems*, 31(6):82–88, 2016. doi: 10.1109/MIS.2016.94.

Amir Zadeh, Paul Pu Liang, Soujanya Poria, Erik Cambria, and Louis-Philippe Morency. Multimodal language analysis in the wild: CMU-MOSEI dataset and interpretable dynamic fusion graph. In *Proceedings of the 56th Annual Meeting of the Association for Computational Linguistics*, pp. 2236–2246, 2018. doi: 10.18653/v1/P18-1208.